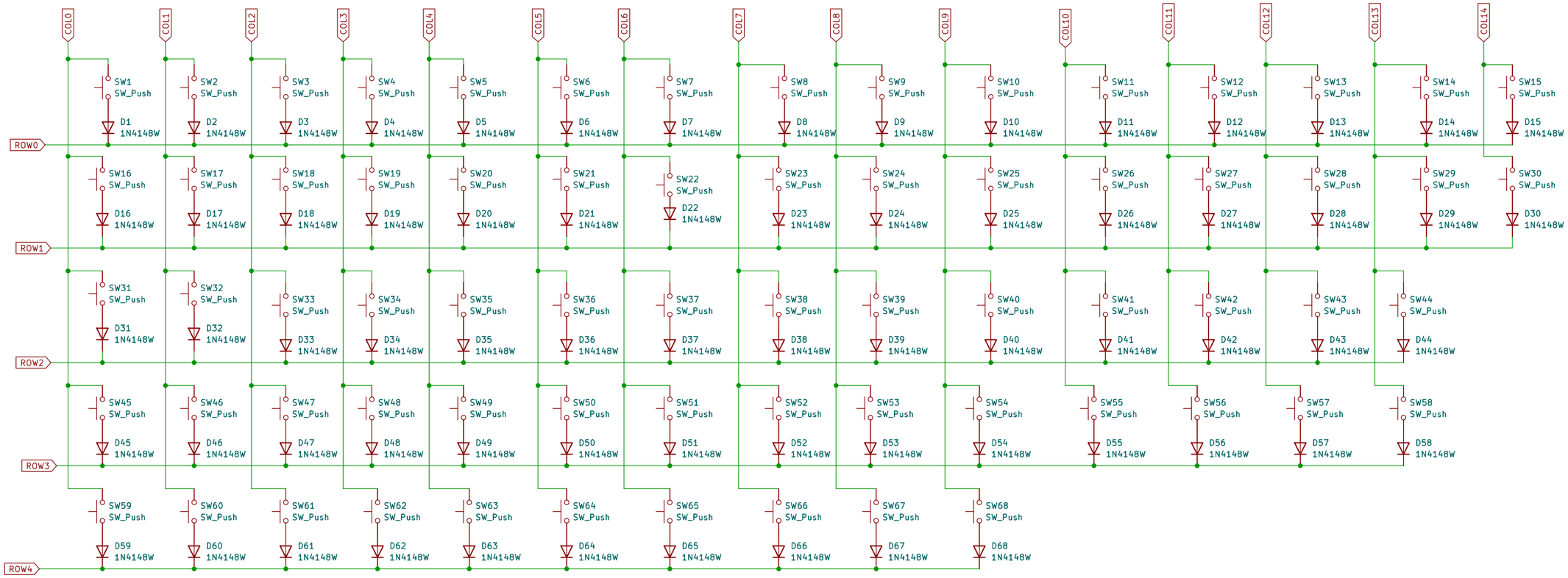




Power_block

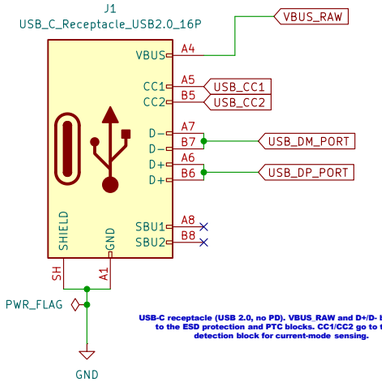
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File: STM H7 KEYBOARD.kicad_sch

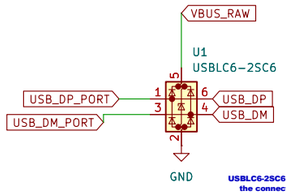
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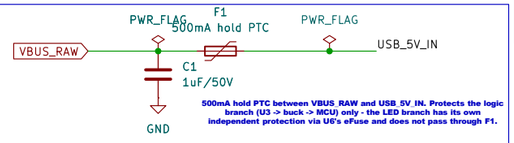
USB PORT



USB-C receptacle (USB 2.0, no PD). VBUS_RAW and D+/D- broken out to the ESD protection and PTC blocks. CC1/CC2 go to the CC detection block for current-mode sensing.

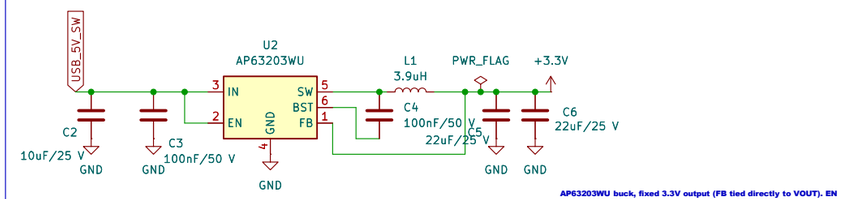


USBLC6-2SC6 TVS array on D+/D- and VBUS_RAW, placed directly after the connector to clamp ESD transients before they reach the rest of the board.



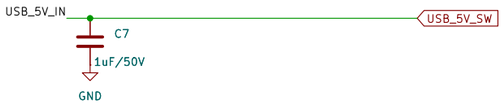
500mA hold PTC between VBUS_RAW and USB_5V_IN. Protects the logic branch (U3 -> buck -> MCU) only - the LED branch has its own independent protection via U6's eFuse and does not pass through F1.

3.3V Power

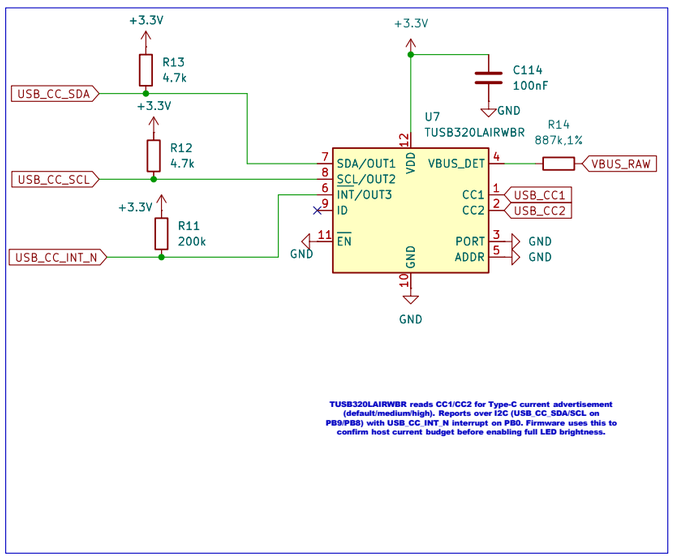


AP63203WU buck, fixed 3.3V output (FB tied directly to VOUT). EN tied to USB_5V_SW for auto-enable. Feeds STM32H723 and all 3.3V logic. Independent of the LED power branch below.

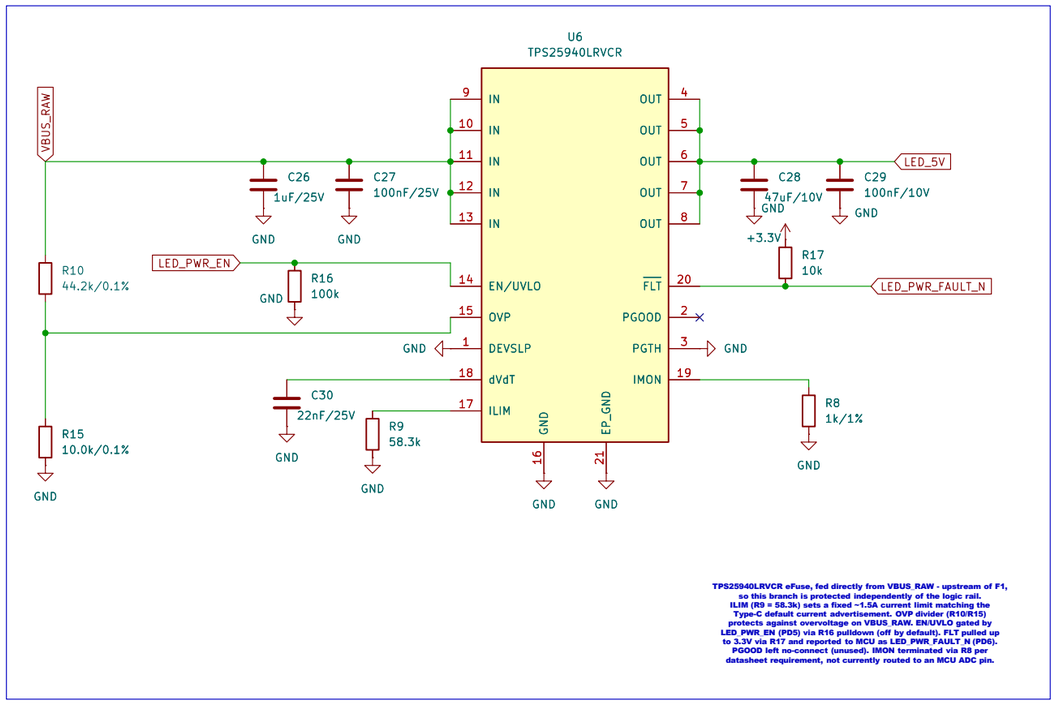
USB POWER BLOCK



PTC output feeds the buck regulator directly. No load switch - removed for simplicity, since the buck has its own internal soft-start.



TUSB320LAIRWBR reads CC1/CC2 for Type-C current advertisement (default/medium/high). Reports over I2C (USB_CC_SDA/SCL on PB9/PB8) with USB_CC_INT_N interrupt on PB0. Firmware uses this to confirm host current budget before enabling full LED brightness.



TPS25940LRVCR eFuse, fed directly from VBUS_RAW - upstream of F1, so this branch is protected independently of the logic rail. ILIM (R9 = 58.3k) sets a fixed ~1.5A current limit matching the Type-C default current advertisement. OVP divider (R10/R15) protects against overvoltage on VBUS_RAW. EN/UVLO gated by LED_PWR_EN (PD5) via R16 pulldown (off by default). FLT pulled up to 3.3V via R17 and reported to MCU as LED_PWR_FAULT_N (PD6). PGOOD left no-connect (unused). IMON terminated via R8 per datasheet requirement, not currently routed to an MCU ADC pin.

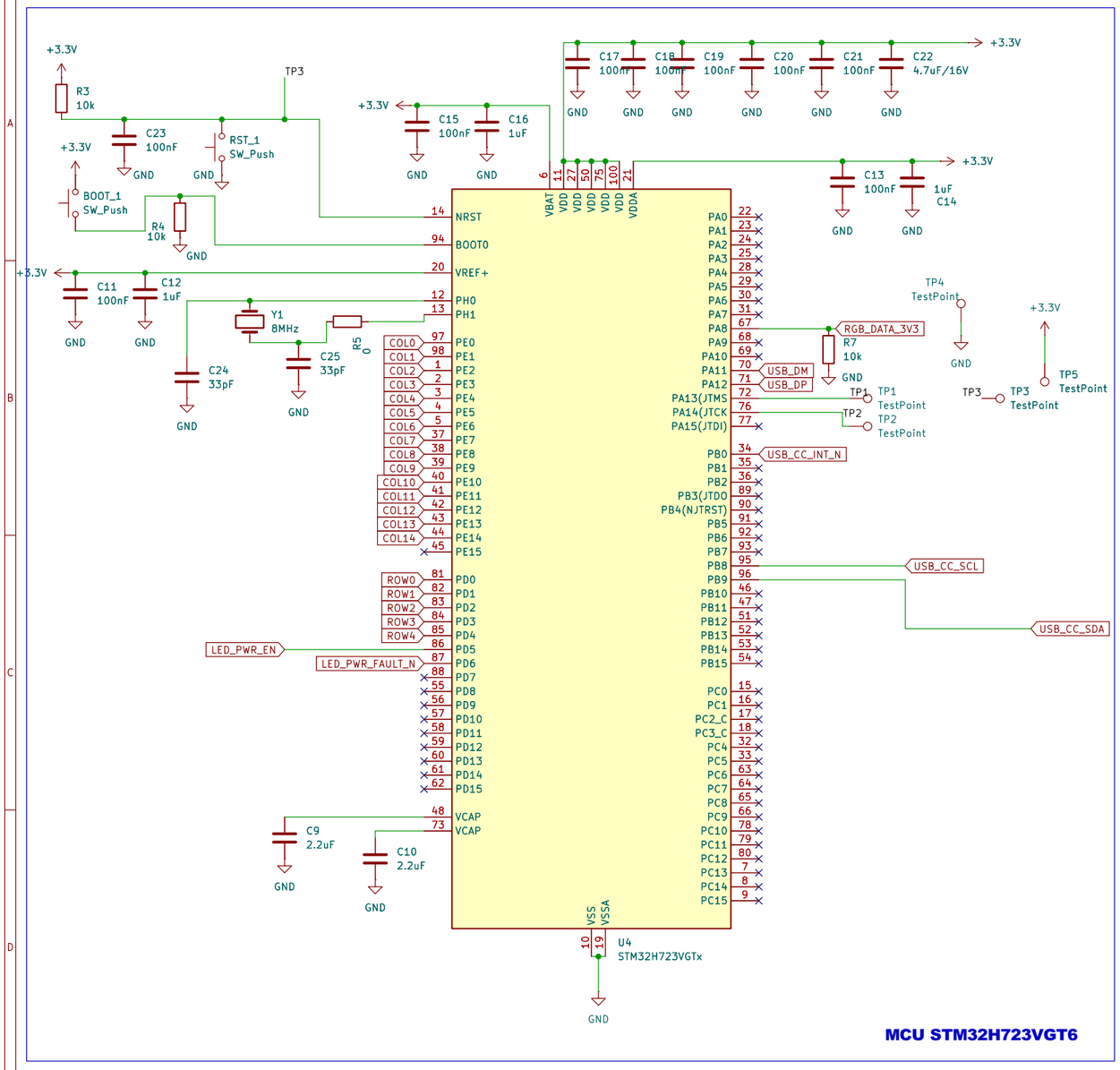
MCU

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RGBSK

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Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/4

Sheet: /Power_block/MCU/RGBSK/
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